

NLP Tutorial: Classification tasks using Deep Learning models

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Important remarks

In order to perform all experiments on the both datasets, you will need to modify the file format of the datasets and adapt the python scripts given as example.

WARNING : In order to run the Transformer script, you shall need a GPU. Otherwise, it should take very long time

1 Tools available

- Two scripts written in python
 - One to use CNN and LSTM (`sentiment_analysis_classification_cnn_lstm.py`)
 - One to use Transformers models (`transformers_classification.py`)
 - Three datasets, use two of them :
 - Massive Multilingual Natural Language Understanding Task (in French – Mandatory)
 - Twitter sentiment data (in English – Optionnal if Allociné is not used)
 - Allociné movie reviews (in French – Optionnal if Twitter is not used)
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2 Work TODO

2.1 Legacy Deep Learning

In order to realise this project, you have to use scripts in python to :

- load the data
- define DL & ML models
- train CNN & LSTM models
- evaluate your models

2.2 Transformers

For the Transformer models, I advise you to use ALBERT models :

- for English : <https://huggingface.co/albert-base-v2>
- for French : <https://huggingface.co/cservan/fralbert-base-cased>
- for Multilingual : <https://huggingface.co/cservan/malbert-base-cased-128k>

Optionally, you can you can use :

- CamemBERT model for French : <https://huggingface.co/almanach/camembert-base>
- Modern CamemBERT model for French (which uses a larger context) :
<https://huggingface.co/almanach/moderncamembert-base>
- multilingual BERT model : <https://huggingface.co/google-bert/bert-base-multilingual-cased>

2.3 Extension

Finally, you have to perform for each task and each models :

- Evaluation with result comparison / explanation / analysis
 - What happend if you apply subword unit tokenization to CNN/LSTM?
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